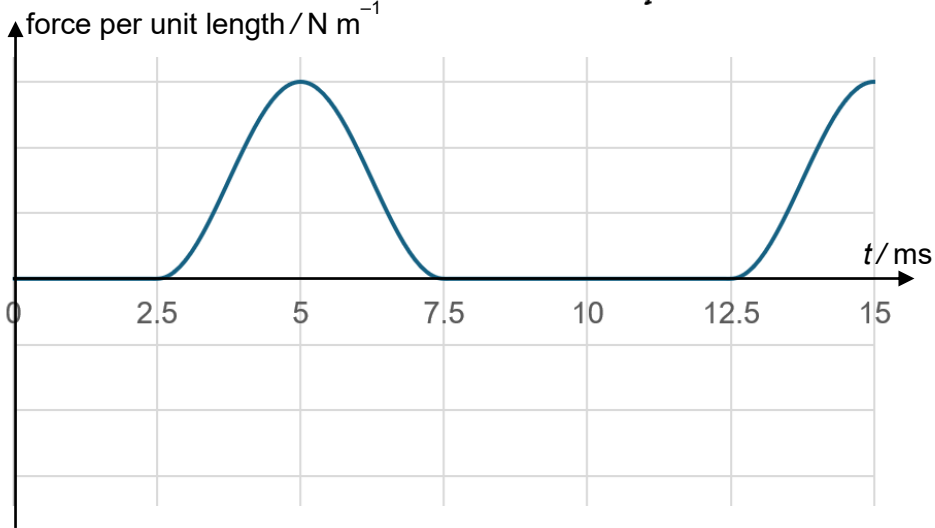


Qns	Answer	Marks
2(a)(i)	$R = \frac{\rho L}{A} = \frac{\rho L}{\pi r^2}$ $= \frac{(1.1 \times 10^{-6})(120 \times 10^{-2})}{\pi(\frac{1.1}{2} \times 10^{-3})^2}$ $= 1.38898... = 1.4 \, \Omega \text{ (2 s.f.)}$	M1
2(a)(ii)	Let potential at Y be 5.0 V $R_{PQ} // 1.8 \, \Omega = \frac{1.4 \times 1.8}{1.4 + 1.8} = 0.7875 \, \Omega$ $\text{PD across } R_{PQ} = \frac{0.7875}{0.7875 + 2.5} \times 5.0 = 1.198 \, \text{V}$	C1
	$\text{PD across XY} = 5.0 - \frac{1.198}{2}$ $= 4.4 \, \text{V}$	C1 A1
2(b)(i)	<u>Current in wire A is in magnetic field produced by current in wire B.</u> Wire A experiences a magnetic force towards wire B according to <u>Fleming's Left Hand Rule.</u>	B1
	From <u>Newton's third law</u>, an equal and opposite force acts on wire B. (or apply FLH rule again on wire B)	B1
2(b)(ii)1.	$I_A = -3.0 \cos(200\pi t) = -3.0 \cos(200\pi(6.5 \times 10^{-3}))$ $= 1.76 \, \text{A}$ From graph, $I_B = 1.2 \, \text{A}$	C1 For both I_A and I_B
	$B_B = \frac{\mu_0 I}{2\pi d}$ $= \frac{(4\pi \times 10^{-7})(1.2)}{2\pi(0.05)} = 4.8 \times 10^{-6}$ $F \text{ per unit length on wire A} = B_B I_A L / L = B_B I_A$ $= (4.8 \times 10^{-6})(1.76)$ $= 8.4 \times 10^{-6} \, \text{N m}^{-1}$	C1
		A1

2(b)(ii)2.		
	• positions and shape of graph during intervals of non-zero force • positions of intervals of zero force	B1 B1
2(b)(ii)3.	$\langle P \rangle = I_{rms}^2 R = \left(\frac{3.0}{2} \right)^2 (15)$ $= 34 \text{ W}$	A1

